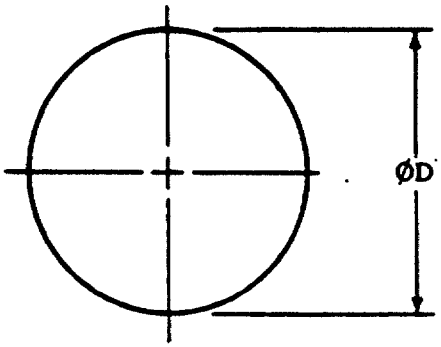
MILITARY SPECIFICATION SHEET
BALLS, BEARING, STEEL, CHROME ALLOY, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bearing described herein shall consist of this specification and the latest issue of DOD-B-82737.



NOTES:

1. All dimensions are in millimeters. Column headings in tolerance tables are in accordance with AFBMA-STD-10.
2. For tolerances by grade see Table II for individual balls and Table III for lots of balls. Tolerances are in micrometers (μm).
3. Grade 10 balls are preferred in most military applications.
4. Referenced government (or nongovernment) documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

FIGURE 1. BALLS, BEARING, STEEL, CHROME ALLOY, METRIC.

FSC 3110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:23Z
Check the source to verify that this is the current version before use.

TABLE 1. Dash numbers and dimensions.

ØD NOM	GRADE 3	GRADE 5	GRADE 10	GRADE 16	GRADE 24	GRADE 48	GRADE 100	GRADE 200
	DASH NUMBERS							
0.3	—	503	1003	1603	2403	—	—	—
0.4	—	504	1004	1604	2404	—	—	—
0.5	—	505	1005	1605	2405	—	—	—
0.6	—	506	1006	1606	2406	—	—	—
0.7	—	507	1007	1607	2407	—	—	—
0.8	308	508	1008	1608	2408	4808	10008	20008
1	310	510	1010	1610	2410	4810	10010	20010
1.2	312	512	1012	1612	2412	4812	10012	20012
1.5	315	515	1015	1615	2415	4815	10015	20015
2	320	520	1020	1620	2420	4820	10020	20020
2.5	325	525	1025	1625	2425	4825	10025	20025
3	330	530	1030	1630	2430	4830	10030	20030
3.5	335	535	1035	1635	2435	4835	10035	20035
4	340	540	1040	1640	2440	4840	10040	20040
4.5	345	545	1045	1645	2445	4845	10045	20045
5	350	550	1050	1650	2450	4850	10050	20050
5.5	355	555	1055	1655	2455	4855	10055	20055
6	360	560	1060	1660	2460	4860	10060	20060
6.5	365	565	1065	1665	2465	4865	10065	20065
7	370	570	1070	1670	2470	4870	10070	20070
7.5	375	575	1075	1675	2475	4875	10075	20075
8	380	580	1080	1680	2480	4880	10080	20080
8.5	385	585	1085	1685	2485	4885	10085	20085
9	390	590	1090	1690	2490	4890	10090	20090
10	3100	5100	10100	16100	24100	48100	100100	200100
11	3110	5110	10110	16110	24110	48110	100110	200110
11.5	3115	5115	10115	16115	24115	48115	100115	200115
12	3120	5120	10120	16120	24120	48120	100120	200120
13	3130	5130	10130	16130	24130	48130	100130	200130
14	3140	5140	10140	16140	24140	48140	100140	200140
15	3150	5150	10150	16150	24150	48150	100150	200150
16	3160	5160	10160	16160	24160	48160	100160	200160
17	3170	5170	10170	16170	24170	48170	100170	200170
18	3180	5180	10180	16180	24180	48180	100180	200180
19	3190	5190	10190	16190	24190	48190	100190	200190
20	3200	5200	10200	16200	24200	48200	100200	200200
21	3210	5210	10210	16210	24210	48210	100210	200210
22	3220	5220	10220	16220	24220	48220	100220	200220
23	3230	5230	10230	16230	24230	48230	100230	200230
24	3240	5240	10240	16240	24240	48240	100240	200240
25	3250	5250	10250	16250	24250	48250	100250	200250
26	—	5260	10260	16260	24260	48260	100260	200260

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:23Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and dimensions. - Continued

ØD NOM	GRADE 3	GRADE 5	GRADE 10	GRADE 16	GRADE 24	GRADE 48	GRADE 100	GRADE 200
	DASH NUMBERS							
28	—	5280	10280	16280	24280	48280	100280	200280
30	—	5300	10300	16300	24300	48300	100300	200300
32	—	5320	10320	16320	24320	48320	100320	200320
34	—	5340	10340	16340	24340	48340	100340	200340
35	—	5350	10350	16350	24350	48350	100350	200350
36	—	5360	10360	16360	24360	48360	100360	200360
38	—	5380	10380	16380	24380	48380	100380	200380
40	—	—	—	—	—	48400	100400	200400
45	—	—	—	—	—	48450	100450	200450
50	—	—	—	—	—	48500	100500	200500
55	—	—	—	—	—	48550	100550	200550
60	—	—	—	—	—	48600	100600	200600
65	—	—	—	—	—	48650	100650	200650

REQUIREMENTS:

1. Material: Steel, chrome alloy conforming to the chemical composition of UNS G51986 in accordance with AMS 6449 or UNS 52986 in accordance with AMS 6440 and as specified in procurement document.

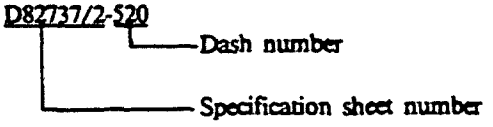
2. Hardness: Shall be 60-67 HRC or equivalent.

3. Surface Roughness: Tolerance limits as specified in Table II and in accordance with ANSI B46.1.

4. Material Density: Shall be 7.833 grams per cubic centimeter.

5. Part Number: Consists of the letter "D", the basic number of this specification sheet and a dash number taken from Table I.

Example: D82737/2-520



D82737/2-520 Indicates a grade 5 bearing ball of chrome alloy steel with a basic diameter of 2 millimeters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:23Z
Check the source to verify that this is the current version before use.

TABLE II. Tolerances by grade for individual balls.

GRADE	ALLOWABLE BALL DIAMETER VARIATION 1/	ALLOWABLE DEVIATION FROM SPHERICAL FORM 1/	MAXIMUM SURFACE ROUGHNESS ARITHMETICAL AVERAGE 1/
	V_D	W	
3	0.08	0.08	0.012
5	0.13	0.13	0.02
10	0.25	0.25	0.025
16	0.4	0.4	0.025
24	0.6	0.6	0.05
48	1.2	1.2	0.08
100	2.5	2.5	0.125
200	5	5	0.2

Tolerances in micrometers (μm)

TABLE III. Tolerances by grade for lots of balls.

GRADE	ALLOWABLE LOT DIAMETER VARIATION 1/	BASIC DIAMETER TOLERANCE 1/	ALLOWABLE BALL GAGE DEVIATION 1/		CONTAINER MARKING INCREMENT 1/
			HIGH	LOW	
3	0.13	±0.75	+0.75	−0.75	0.25
5	0.25	+1.25	+1.25	−1	0.25
10	0.5	±2.5	+1.25	−1	0.25
16	0.8	±2.5	+1.25	−1	0.25
24	1.2	±2.5	+2.5	−2.5	0.25
48	2.4	±5	—2/	—2/	1.25
100	5	±12.5	—2/	—2/	—2/
200	10	±25	—2/	—2	—2/

1/ Tolerances in micrometers (μm)
2/ Not applicable

Custodians:
Army - AT
Air Force - 99

Review activities:
Navy - SH
Air Force - 84
DLA - IS

Preparing activity:
Navy - OS

Agent:
DLA - IS

(Project 3110-0737-02)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:23Z
Check the source to verify that this is the current version before use.